

K L UNIVERSITY
FRESHMAN ENGINEERING DEPARTMENT
A Project Based Lab Report
On
Automated Household Inventory
Intelligent Consumption Analytics System

SUBMITTED BY:

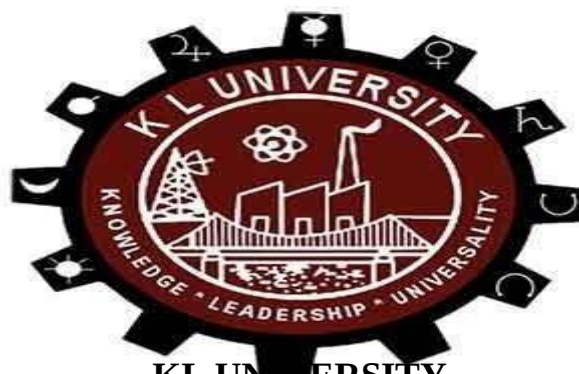
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CERTIFICATE

This is to certify that the project based laboratory report entitled “**Automated Household Inventory Intelligent Consumption Analytics System**” submitted by **BONAM SANJAY KRISHNA** bearing Regd. No. **2500040173** to the **Department of COMPUTER SCIENCE ENGINEERING , KL University** in partial fulfillment of the requirements for the completion of a project based Laboratory in “Computational Thinking for a project based Laboratory (25SC2007)” course in I B Tech II Semester, is a Bonafede record of the work carried out by him under my supervision during the academic year 2025 – 2026.

PROJECT SUPERVISOR

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Finally, we are pleased to acknowledge our indebtedness to all those who devoted themselves directly or indirectly to making this project a success.

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ABSTRACT

The Automated Household Inventory & Intelligent Consumption Analytics System (HI-CAS) is a Python-based application designed to automate and optimize household inventory management. Many households rely on manual tracking methods such as memory or handwritten lists, which often lead to over-purchasing, stock shortages, expired items, and unnecessary financial loss. HI-CAS addresses these issues by providing a structured and intelligent system for managing daily household items efficiently.

The system enables users to add, update, delete, and monitor inventory items while tracking quantities and expiry dates. It generates low-stock alerts and expiry notifications to prevent shortages and reduce wastage. In addition, the system performs basic consumption analytics to analyze usage patterns and support better purchasing decisions.

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1.INTRODUCTION

Automated Household Inventory & Intelligent Consumption Analytics System (HI-CAS) The Automated Household Inventory & Intelligent Consumption Analytics System (HI-CAS) is a Python-based software application designed to simplify and optimize the management of household inventory. In modern homes, managing groceries, kitchen supplies, cleaning materials, and daily consumables is often done manually using memory, handwritten lists, or simple notes. This traditional approach frequently results in over-purchasing, under-stocking of essential items, expired products, and unnecessary financial waste. HI-CAS aims to solve these problems by providing an automated, structured, and intelligent inventory management system. The system records item details such as name, quantity, category, and expiry date, while also tracking consumption patterns over time. By analyzing usage trends, the system generates smart alerts for low stock and upcoming expiries, helping households make informed purchasing decisions.

2.AIM :

Automated Household Inventory & Intelligent Consumption Analytics System (HI-CAS)

In many households, inventory management is handled manually through memory, handwritten lists, or basic notes. This unstructured approach often leads to several common problems such as over-purchasing, under-stocking of essential goods, product expiration, food wastage, and poor budgeting. Without a proper tracking mechanism, households lack visibility into what items are available, how frequently they are consumed, and when they need to be replenished.

The absence of a centralized and automated system makes it difficult to:

- Maintain accurate records of household items
- Track item quantities in real time
- Monitor expiry dates of perishable goods
- Analyze consumption patterns over time
- Predict future purchasing needs
- Generate alerts for low-stock items

3.ADVANTAGES

- **Efficiency:** Automates tracking of household items, reducing reliance on memory or manual lists.
- **Cost Savings:** Prevents over-purchasing and reduces wastage by alerting users about low stock and expiring items.
- **Organization:** Provides a structured system to manage inventory, making it easier to know what's available at any time.
- **Analytics Insight:** Tracks consumption patterns, helping households make smarter purchasing decisions.
- **Scalability:** Can be expanded with features like barcode scanning, mobile sync, or IoT integration (smart kitchen sensors).
- **User-Friendly Alerts:** Timely notifications prevent shortages and expired products, improving convenience.

DISADVANTAGES

- **Initial Setup Effort:** Users must input all items manually at first, which can be time-consuming.
- **Maintenance:** Requires regular updates (adding/removing items) to stay accurate — if users neglect this, the system loses reliability.
- **Technical Barriers:** Some households may find it difficult to use if they're not comfortable with apps or digital tools.
- **Data Accuracy Dependence:** Analytics are only as good as the data entered; incorrect entries can lead to misleading insights.
- **Resource Usage:** Running on Python with databases may require a device always available (PC, Raspberry Pi, or server), which adds overhead.
- **Limited Context Awareness:** Without IoT integration, the system can't automatically detect consumption — it relies on manual updates.

4.FUTURE IMPLEMENTATION

Although the HI-CAS system provides automated inventory management and basic consumption analytics, it can be further enhanced to make it more intelligent, scalable, and suitable for real-world deployment. The following improvements can be implemented in future versions:

Database Integration

- Support large-scale data handling
- Improve data security and reliability
- Enable faster data retrieval and multi-user access

Web or Mobile Application

- Convert the system into a web app using Flask / Django
- Develop a mobile app version (Android/iOS)
- Enable remote access and synchronization
- Provide cloud-based storage and backup

AI-Based Consumption Prediction

- Use Machine Learning algorithms to predict future consumption
- Suggest optimal purchasing quantities
- Detect unusual usage patterns
- Provide smart budget planning recommendations

Barcode / QR Code Integration

- Scan product barcodes for quick entry
- Automatically fetch product details
- Reduce manual input errors

Notification System

- Email or SMS alerts for low stock and expiry
- Push notifications in mobile version
- Reminder system for monthly shopping

5.SYSTEM REQUIREMENTS

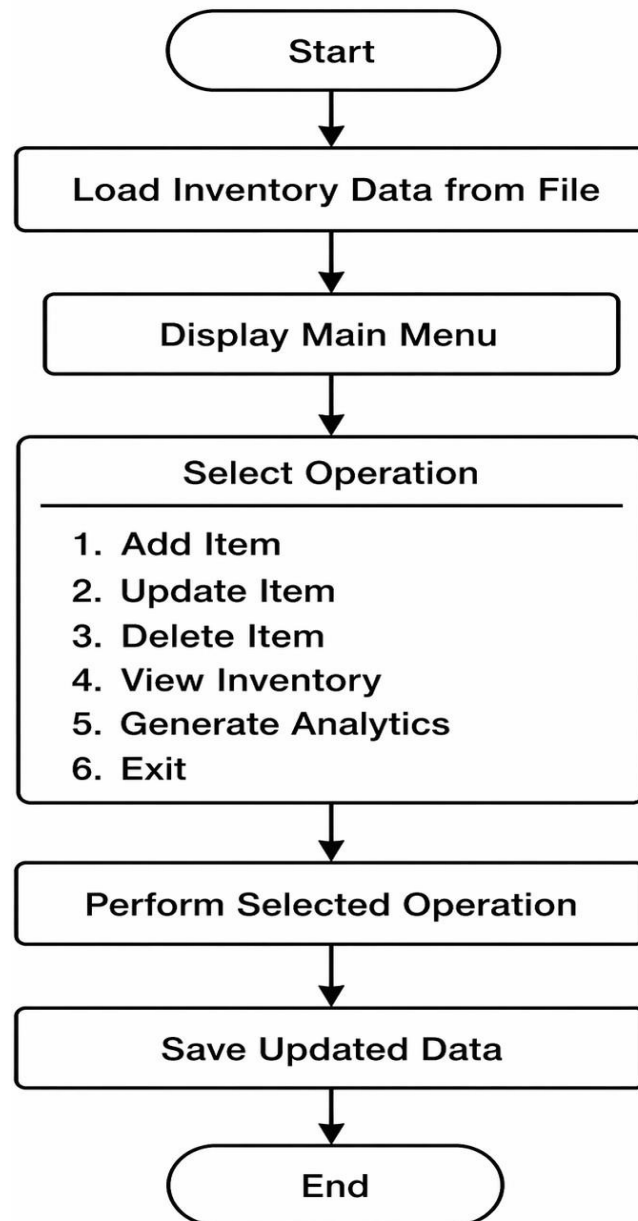
➤ SOFTWARE REQUIREMENTS:

- Operating System: Windows / Linux / macOS
- Python 3.x
- IDE: VS Code / PyCharm / IDLE
- Required Libraries:
 - csv (for file handling – built-in)

➤ HARDWARE REQUIREMENTS:

- Computer / Laptop
- Minimum 4 GB RAM
- 20 GB free storage (minimum)
- Keyboard & Monitor

6.FLOW CHART DIAGRAM :



7.ALGORITHM

Step 1 : START

Step 2 : Load data from CSV file

Step 3 : Show Menu

1. Add Item
2. Update Item
3. Delete Item
4. View Items
5. Low Stock Alert
6. Expiry Alert
7. Exit

Step 4 : Read Choice

Step 5 : IF 1 → Read Name, Qty, Expiry → Save

IF 2 → Read Name, New Qty → Update

IF 3 → Read Name → Delete

IF 4 → Display all Items

IF 5 → Show items where $Qty \leq 2$

IF 6 → Show items where $Expiry \leq \text{Today}$

IF 7 → Save to CSV → STOP

ELSE → Print "Invalid Choice"

Step 6 : REPEAT from Step 3

8.IMPLEMENTATION

class Item:

```
def __init__(self, name, quantity, expiry):
    self.name = name
    self.quantity = int(quantity)
    self.expiry = expiry # format: YYYY-MM-DD

def is_expired(self):
    today = datetime.today().date()
    expiry_date = datetime.strptime(self.expiry, "%Y-%m-%d").date()
    return expiry_date <= today

def __str__(self):
    return f"{self.name} | Qty: {self.quantity} | Expiry: {self.expiry}"
```

----- Inventory Class -----

class Inventory:

```
def __init__(self, file_name="inventory.csv"):
    self.file_name = file_name
    self.items = {}
    self.load_data()

# Load data from CSV
def load_data(self):
    try:
        with open(self.file_name, mode='r') as file:
            reader = csv.reader(file)
            for row in reader:
                name, qty, expiry = row
                self.items[name] = Item(name, qty, expiry)
    except FileNotFoundError:
        pass
```

Save data to CSV

```
def save_data(self):
    with open(self.file_name, mode='w', newline='') as file:
        writer = csv.writer(file)
        for item in self.items.values():
            writer.writerow([item.name, item.quantity, item.expiry])
```

Add Item

```
def add_item(self, name, quantity, expiry): 14
```

```

self.items[name] = Item(name, quantity, expiry)
print("Item added successfully!")

# Update Item
def update_item(self, name, quantity):
    if name in self.items:
        self.items[name].quantity = int(quantity)
        print("Item updated successfully!")
    else:
        print("Item not found!")

# Delete Item
def delete_item(self, name):
    if name in self.items:
        del self.items[name]
        print("Item deleted successfully!")
    else:
        print("Item not found!")

# Display Inventory
def display_items(self):
    if not self.items:
        print("Inventory is empty.")
    for item in self.items.values():
        print(item)

# Low Stock Alert
def low_stock_alert(self, threshold=2):
    print("\nLow Stock Items:")
    for item in self.items.values():
        if item.quantity <= threshold:
            print(item.name)

# Expiry Alert
def expiry_alert(self):
    print("\nExpired / Expiring Items:")
    for item in self.items.values():
        if item.is_expired():
            print(item.name)

# Consumption Analytics (Simple Example)
def total_items(self):
    total = sum(item.quantity for item in self.items.values())
    print(f"\nTotal Quantity of All Items: {total}")

```

----- Main Program -----

```
def main():
    inventory = Inventory()

    while True:
        print("\n==== HI-CAS MENU ====")
        print("1. Add Item")
        print("2. Update Item")
        print("3. Delete Item")
        print("4. View Inventory")
        print("5. Low Stock Alert")
        print("6. Expiry Alert")
        print("7. Total Quantity Report")
        print("8. Exit")

        choice = input("Enter choice: ")

        if choice == "1":
            name = input("Enter item name: ")
            qty = input("Enter quantity: ")
            expiry = input("Enter expiry date (YYYY-MM-DD): ")
            inventory.add_item(name, qty, expiry)

        elif choice == "2":
            name = input("Enter item name to update: ")
            qty = input("Enter new quantity: ")
            inventory.update_item(name, qty)

        elif choice == "3":
            name = input("Enter item name to delete: ")
            inventory.delete_item(name)

        elif choice == "4":
            inventory.display_items()

        elif choice == "5":
            inventory.low_stock_alert()

        elif choice == "6":
            inventory.expiry_alert()

        elif choice == "7":
            inventory.total_items()

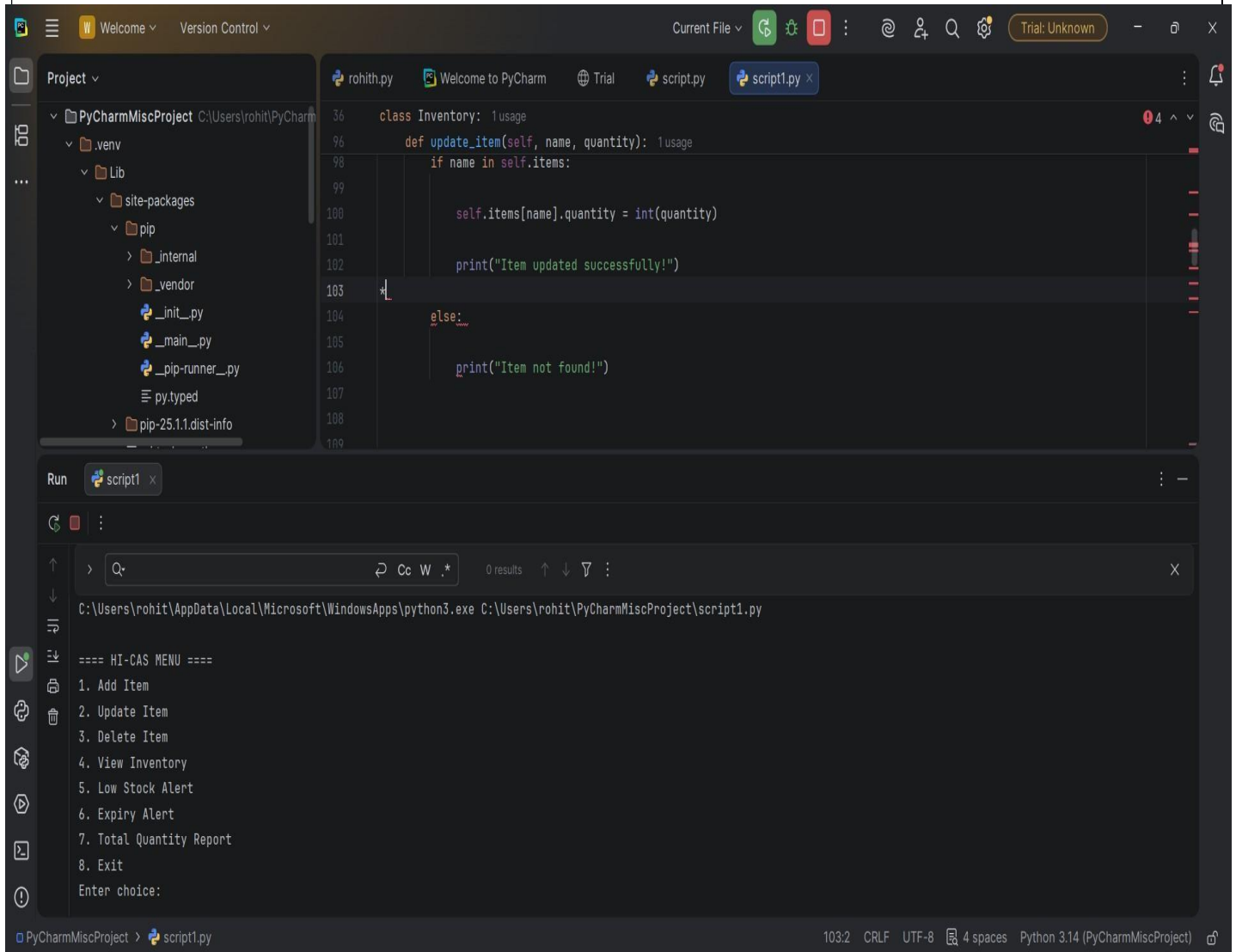
        elif choice == "8":
```

```
inventory.save_data()
print("Data saved. Exiting program.")
break
```

```
else:
    print("Invalid choice! Try again.")
```

```
if __name__ == "__main__":
    main()
```


screen short :



```
Run PythonProject x
C:\Users\rohit\AppData\Local\Microsoft\WindowsApps\python3.exe C:\Users\rohit\PyCharmMiscProject\PythonProject.py

==== HI-CAS MENU ====
1. Add Item
2. Update Item
3. Delete Item
4. View Inventory
5. Low Stock Alert
6. Expiry Alert
7. Total Quantity Report
8. Exit
Enter choice: 1
Enter item name: soap
Enter quantity: 2
Enter expiry date (YYYY-MM-DD): 2026-08-16
Item added successfully!

==== HI-CAS MENU ====
1. Add Item
2. Update Item
3. Delete Item
4. View Inventory
5. Low Stock Alert
6. Expiry Alert
7. Total Quantity Report
8. Exit
Enter choice: 4
ertyu | Qty: 3 | Expiry: 3333-33-33
soap | Qty: 2 | Expiry: 2026-08-16

PyCharmMiscProject > PythonProject.py 149:11 CRLF UTF-8 4 spaces Python 3.14 (PyCharmMiscP
```

Output :

1. Add Item
2. Update Item
3. Delete Item
4. View Inventory
5. Low Stock Alert
6. Expiry Alert
7. Total Quantity Report
8. Exit

Enter choice: 1

Enter item name: Rice

Enter quantity: 5

Enter expiry date (YYYY-MM-DD): 2026-03-01

Item added successfully!

Enter choice: 4

Rice | Qty: 5 | Expiry: 2026-03-01

10.Conclusion :

The Automated Household Inventory & Intelligent Consumption Analytics System (HI-CAS) successfully demonstrates how Python can be used to design a structured, intelligent, and automated inventory management solution for households. The system replaces manual tracking methods with a digital, organized, and data-driven approach that improves efficiency and reduces wastage.

Through the implementation of Object-Oriented Programming (OOP) principles such as Encapsulation, Abstraction, Inheritance, and Polymorphism, the project maintains a modular and scalable architecture. The use of file handling ensures persistent data storage, while built-in analytics features provide insights into consumption patterns, stock levels, and expiry management.

The system effectively supports:

- Automated item management
- Real-time stock monitoring
- Low-stock and expiry alerts
- Basic consumption analytics
- Structured data storage

By integrating automation with analytical insights, HI-CAS helps users make smarter purchasing decisions, optimize household budgeting, and minimize unnecessary waste.

Overall, this project not only serves as a practical household management tool but also demonstrates strong software engineering concepts, making it a valuable academic project with potential for real-world expansion through future enhancements such as database integration, GUI development, cloud deployment, and AI-based consumption prediction.